Fiscal Year:	FY 2006	**Task Last Updated:**	FY 11/17/2006
PI Name:	Nabity, James A. Ph.D.		
Project Title:	A Lightweight Freeze-Tolerant Radiator for an EMU		
Division Name:	Human Research		
Program/Discipline:	ADVANCED HUMAN SUPPORT TECHNOLOGIES		
Program/Discipline--Element/Subdiscipline:	ADVANCED HUMAN SUPPORT TECHNOLOGIES--Advanced extravehicular activity		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
PI Email:	nabity@tda.com	**Fax:**	FY 303-422-7763
PI Organization Type:	INDUSTRY	**Phone:**	303-940-2313
Organization Name:	TDA Research, Inc.		
PI Address 1:	12345 W. 52nd Ave		
PI Address 2:			
PI Web Page:			
City:	Wheat Ridge	**State:**	CO
Zip Code:	80033-1917	**Congressional District:**	7
Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	2002 Space Biology 02-OBPR-01
Start Date:	08/23/2003	**End Date:**	08/31/2006
No. of Post Docs:	0	**No. of PhD Degrees:**	0
No. of PhD Candidates:	0	**No. of Master' Degrees:**	0
No. of Master's Candidates:	0	**No. of Bachelor's Degrees:**	0
No. of Bachelor's Candidates:	0	**Monitoring Center:**	NASA JSC
Contact Monitor:	Trevino, Luis	**Contact Phone:**	(281) 483-9141
Contact Email:	luis.trevino-1@nasa.gov		
Flight Program:			
Flight Assignment:			
Key Personnel Changes/Previous PI:	0		
COI Name (Institution):			
Grant/Contract No.:	NAS 9-03052		
Performance Goal No.:			
Performance Goal Text:			

During an ExtraVehicular Activity (EVA), both the heat generated by the astronaut’s metabolism and that produced by the Portable Life Support System (PLSS) must be rejected to space. The heat sources include the heat of adsorption of metabolic CO2, the heat of condensation of water, the heat removed from the body by the liquid cooling garment and the load from the electrical components. Although the sublimator hardware to reject this load weighs only 3.48 lbs, an additional eight pounds of water are loaded into the unit of which about six to eight are sublimated and lost (or expended); this is the single largest expendable during an eight-hour EVA. While losing eight pounds of water per EVA has been acceptable for low earth orbit missions where resupply is possible, albeit expensive (~$80,000/EVA), it is not acceptable for moon and interplanetary missions where resupply will be difficult or impossible.
Using a radiator to reject heat from the Astronaut during an EVA, we can significantly reduce the amount of expendable water consumed by the sublimator, the current Extravehicular Mobility Unit (EMU) thermal control system. However, past systems to reduce the expendables have weighed too much, since heavy-wall tube construction was used to prevent

Task Description:	lines from bursting at low load conditions when freezing can be a problem and thick-walled radiative fins were needed to transfer high heat loads. Thus, a key technological innovation in this project was to develop a lightweight and freeze tolerant heat rejection system to accommodate a variable heat load. This novel heat rejection system used a freeze tolerant heat exchanger and refrigerant loop heat pipes to transfer heat from the cooling water loop to the lightweight fin that radiated the heat to the environment. The basic premise is that we can make a large reduction in system weight if only the small heat exchanger must be freeze-tolerant (instead of the much larger radiator). During the execution of the project we designed, built and tested critical components, such as a freeze tolerant heat exchanger, a loop heat pipes and radiator coupons, to assess our designs. We successfully demonstrated that our freeze tolerant heat exchanger could be turned down by freezing and could be cycled between high and low heat rejection rates without failure. We also developed numerical models to simulate the water flow through the heat exchanger during freeze and thaw scenarios. The simulations showed us that we could passively control which water channels freeze and therefore, also to passively control the heat rejection. We first built a single loop heat pipe (evaporator and condensing tube), bonded it to a carbon fiber radiator fin and then tested it in a simulated lunar environment. Robust and stable heat rejection performance was demonstrated for a broad range of operating conditions. We also conducted impact resistance testing on carbon fiber and aluminum radiator fin materials. The test results showed aluminum to be a better choice, since the brittle carbon fiber composites broke too easily. Therefore, TDA built the prototype radiator and loop heat pipes from aluminum. The freeze tolerant heat exchanger was fabricated from copper tubes. Both MLI and Aspen SpaceLoft AR3103 aerogel insulations were installed. The resulting prototype was designated the Radiator And Freeze Tolerant heat eXchanger (RAFT-X). The RAFT-X weighed 19.0 lbm when fully assembled and charged with refrigerant. The heat transfer performance was measured in tests conducted at TDA and NASA JSC. In tests at TDA we were able to vary the heat rejection rate from 170 to 765 Btu per hour. In tests at the NASA JSC Chamber E Thermal Vacuum Facility, more than 700 Btu/h of heat were rejected in lunar and Mars environments with temperatures as cold as –150 F and it would have been about 20% greater had the desired chamber temperature of -250°F been achieved. Further, the system was cycled through three complete freeze / thaw cycles without damage to the heat exchanger. We were also able to operate the heat exchanger in a partially frozen configuration to throttle the heat rejection rate from 530 Btu/h at low water flow rate down to 300 Btu/h. Finally, the deliberate loss of a single loop heat pipe only degraded the heat rejection performance by about 2 to 5%. These test results conclusively show the robust heat rejection performance that the RAFT-X concept can provide to an astronaut during EVA.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	The largest and nearest term commercial applications are the use of freeze tolerant tubing on earth. These earth-based applications include sprinkler systems and potable water supply in homes and commercial buildings. This market is potentially very large and virtually un-tapped because of the lack of a viable freeze tolerant tube. The Insurance Institute for Property Loss Reduction, says frozen pipes cost the insurance companies in the USA $4.2 billion in damage to insured homes and buildings over the past decade (i.e., about $400,000,000 per year). The savings in insurance rates alone could more than offset the cost to the user, who would have the added benefit of not having non-replaceable photographs and other objects destroyed by water damage and their lives disturbed during repairs of the water damage.
	During an ExtraVehicular Activity (EVA), both the heat generated by the astronaut’s metabolism and that produced by the Portable Life Support System (PLSS) must be rejected to space. The heat sources include the heat of adsorption of metabolic CO2, the heat of condensation of water, the heat removed from the body by the liquid cooling garment and the load from the electrical components. Although the sublimator hardware to reject this load weighs only 3.48 lbs, an additional eight pounds of water are loaded into the unit of which about six to eight are sublimated and lost (or expended); this is the single largest expendable during an eight-hour EVA. While losing eight pounds of water per EVA has been acceptable for low earth orbit missions where resupply is possible, albeit expensive (~$80,000/EVA), it is not acceptable for moon and interplanetary missions where resupply will be difficult or impossible. If most of this heat load could be radiated to space, then the amount of water lost to sublimation would be greatly reduced. The current PLSS has enough surface area to radiate roughly 75% of the average heat load. Unfortunately, a radiator tends to reject heat at a relatively constant rate, whereas the astronaut actually produces heat at a rate that can vary from 250 to 2500 Btu/hr, depending on the workload. Without a way to adjust the heat removal rate, the astronaut would alternately suffer from both frostbite and heat stroke. Therefore, in this project TDA Research, Inc. (TDA) developed a heat exchanger to proportionally control the heat rejection rate to match the load. This heat exchanger became an integral part of our Radiator and Freeze Tolerant Heat Exchanger or RAFT-X prototype that we delivered to NASA JSC in 2006. It has long been known that a radiator can be “turned-down” by sequentially allowing tubes that carry the water to the radiator to freeze. When a tube freezes, the temperature of the radiating area around it drops, and the heat rejection rate drops as well. The problem with the freezable radiators developed to date is that they are far too heavy, since they used heavy walled tubes to prevent bursting during freeze/thaw cycles. Further, in past freezable designs, the entire surface of the PLSS had to be thick enough to transfer enough heat to melt the frozen tubes; the radiator must be thick because the heat had to be conducted several inches between adjacent tubes. Thus, we need a lightweight radiator that does not burst when it freezes. In our design, we separated the thermal system into two fluid loops; a water loop and refrigerant loaded loop heat pipes. The water loop picks up heat from the astronaut, the life support system and the electronics. The water loop transfers the heat through a heat exchanger to the loop heat pipes that are bonded to the PLSS shell. The heat pipes distribute heat to the shell, which then radiates the heat to space. The key to our design is the small freezable heat exchanger that we use to couple the water loop to the heat pipes. As the heat load drops during periods when the astronaut is not working hard, some of the heat exchanger flow passages freeze, thereby preventing heat from reaching some of the heat pipes (that do not have freeze/thaw problems) and thereby deactivating portions of the radiator surface. This allows us to use a much lighter radiator shell, since we don’t need the very high lateral conductivity necessary to thaw adjacent flow passages (since they never freeze). For these reasons TDA developed a new freezable heat exchanger that accommodated the expansion of the water when it freezes and returns to its original shape as it thaws. Designing for passive restart or heat exchanger thaw from a fully frozen condition has been particularly difficult, but our design keeps the materials within their elastic limits so that large

Task Progress:	volume changes can be accommodated without breaking the heat exchanger. Although a freezable head exchanger is somewhat heavier than one that does not need to withstand freeze / thaw cycles, there is little weight penalty because the heat exchanger is only a small part of the overall system. In the RAFT-X design, the heat from the astronaut and equipment is transferred through the heat exchanger to the radiator using loop heat pipes. The aluminum PLSS shell rejects heat by radiation to space, and thereby reduces the loss of water to sublimation during an EVA by 3.6 lbs. Further, the colder the environment the greater the savings. Proportional control of the heat rejection rate is achieved by freezing out one or more of the heat exchanger water flow channels. Under low metabolic load, the control valve reduces the water flow to the heat exchanger. The temperature on the waterside of the heat exchanger drops and the channels begin to freeze, starting with the flow channels next to the evaporators. The frozen region progresses inwards as the load continues to drop. At the minimum heat load or lowest water flow rate all flow channels within the heat exchanger may entirely freeze; in this scenario only an isolated bypass tube will remain unfrozen to reject heat to the sublimator. Once the heat load increases, the heat exchanger will thaw to again reject heat to the environment. While past low expendable systems have been too heavy, we estimate our heat exchanger/loop heat pipe and small sublimator water tank to be no heavier than the current system. During development, we fabricated and tested key components of the thermal control system, including a freeze tolerant heat exchanger, loop heat pipes and a carbon fiber composite radiator. An early heat exchanger design was cycled repeatedly through the freeze-trap condition without breaking, demonstrating the viability of our approach. The heat exchanger elastically yielded during freeze as desired and then returned to its original state during thaw. We also conducted extensive modeling of our system; both a SINDA/FLUINT heat exchanger thermal performance model and fully coupled CFD / thermal models were developed to optimize our design. Although the SINDA simulations provided significant insight into the heat exchanger operation, the model proved too sensitive to small perturbations in geometry, material properties and operating conditions. Therefore, we validated a fully coupled computational fluid dynamics and thermal analysis model using propylene glycol / water heat exchanger test data. The model and analysis procedures developed during the validation effort were then applied to numerically evaluate design perturbations of an improved freeze tolerant heat exchanger in which the propylene glycol was replaced with another refrigerant. We also conducted impact resistance testing on carbon fiber and aluminum radiator fin materials to evaluate their response to high impact loads. Preliminary results showed that carbon fiber composites are too brittle for the PLSS shell. Therefore, a large-scale aluminum PLSS mockup was fabricated and tested using both blunt and sharp-corner probes at static forces up to 400 lbf (simulating a fall on Mars). Less damage occurred to the large “backpack” style radiator mockup than had been observed in earlier tests to smaller coupons. Because of the large unsupported area, some force was expended to elastically move the aluminum skin. In contrast, in the small coupon tests the force was concentrated at a single point where it first caused plastic deformation and then eventually ruptured the aluminum. The loop heat pipe inside the radiator skin was undamaged by a 400 lb impact with the sharp-corner probe, except for a small adhesive delamination at the impact site where the skin was dented. Thus, one would expect the thermal control system to remain operational with minimal loss in performance even with rough handling. The experimental and numerical results were used to design and build a full-scale Radiator and Freeze-Tolerant Heat Exchanger. In performance tests at TDA, the RAFT-X demonstrated total heat rejection from 170 to 765 Btu/h; of which about half was attributed to radiation. The RAFT-X functioned through numerous heat rejection cycles from low to high loads. We throttled the heat rejection by varying the flow rate and freezing up the water flow channel. Also, we were able to thaw and then restart the heat exchanger after a hard freeze. The system was then delivered to NASA Johnson Space Center for test and evaluation. Extensive testing in the NASA JSC Chamber E Thermal Vacuum Facility showed that the RAFT-X performed very well. Up to 700 Btu/h of heat were rejected in lunar and Mars environments with temperatures as cold as –150 F. The RAFT-X endured repeated freeze / thaw cycles and in fact, the heat exchanger was completely frozen three times without any apparent damage to the unit. We were also able to operate the heat exchanger in a partially frozen configuration to throttle the heat rejection rate from 530 Btu/h at low water flow rate down to 300 Btu/h. Finally, the loss of a single loop heat pipe only degraded the heat rejection performance by about 2 to 5%. These test results conclusively show the robust heat rejection performance that the RAFT-X concept can provide to an astronaut during EVA.
Bibliography Type:	Description: (Last Updated: 11/18/2010)
Abstracts for Journals and Proceedings	Nabity J, Copeland R, Mason G, Trevino L. "Advanced Lightweight Freeze Tolerant Radiator for the EMU." Presented at the Habitation 2006 Conference, Orlando, FL, 5-8 Feb 2006. Habitation 2006;10(3-4):169-70. , Feb-2006
Papers from Meeting Proceedings	Nabity J, Copeland R, Trevino L. "A Lightweight, Freeze Tolerant Radiator for the EMU." Advanced EVA Technical Forum, 15-16 Nov 2005. Advanced EVA Technical Forum, 15-16 Nov 2005. , Nov-2005
Papers from Meeting Proceedings	Copeland R, Elliott J, Weislogel M, Trevino L. "A Lightweight, Freeze Tolerant Radiator for an EMU." Advanced EVA Technical Forum, Houston, TX, 30 June-1 July 2004. Advanced EVA Technical Forum, 2004. , Jun-2004
Papers from Meeting Proceedings	Trevino L, Copeland R, Elliott J, Weislogel M. "Freeze Tolerant Radiator for the EMU." 34th International Conference on Environmental Systems, July 2004. SAE Paper 04ICES-388, Warrendale, PA : SAE International, 2004. , Jul-2004
Papers from Meeting Proceedings	Copeland R, Elliott J, Weislogel M, Trevino L. "A Lightweight, Freeze Tolerant Radiator for an EMU." Habitation Conference, 6 Jan 2004. HLS5 Poster No. 02, Habitation Conference, 2004. , Jan-2004

Papers from Meeting Proceedings	Nabity J, Copeland R, Mason G, Libberton K, Paul H, Trevino L, Stephan R. "Performance Testing of an Advanced Lightweight Freezable Radiator." 36th International Conference on Environmental Systems, Norfolk, VA, July 17-20, 2006. SAE paper 2006-01-2232, Warrendale, PA : SAE International, 2006. , Jul-2006